

# Association Rule Learning — Groceries Dataset

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<sup>1</sup>Unsupervised Learning Methods – Assignment: Association Rule Learning

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**Software used:** Python (pandas, mlxtend – Apriori algorithm)

## Question 1: Understand the Data

- a. **Number of transactions:** 9,835
- b. **Number of unique grocery items:** 169
- c. **Top 10 most frequently purchased items:**

| Rank | Item             | # Transactions |
|------|------------------|----------------|
| 1    | whole milk       | 2,513          |
| 2    | other vegetables | 1,903          |
| 3    | rolls/buns       | 1,809          |
| 4    | soda             | 1,715          |
| 5    | yogurt           | 1,372          |
| 6    | bottled water    | 1,087          |
| 7    | root vegetables  | 1,072          |
| 8    | tropical fruit   | 1,032          |
| 9    | shopping bags    | 969            |
| 10   | sausage          | 924            |

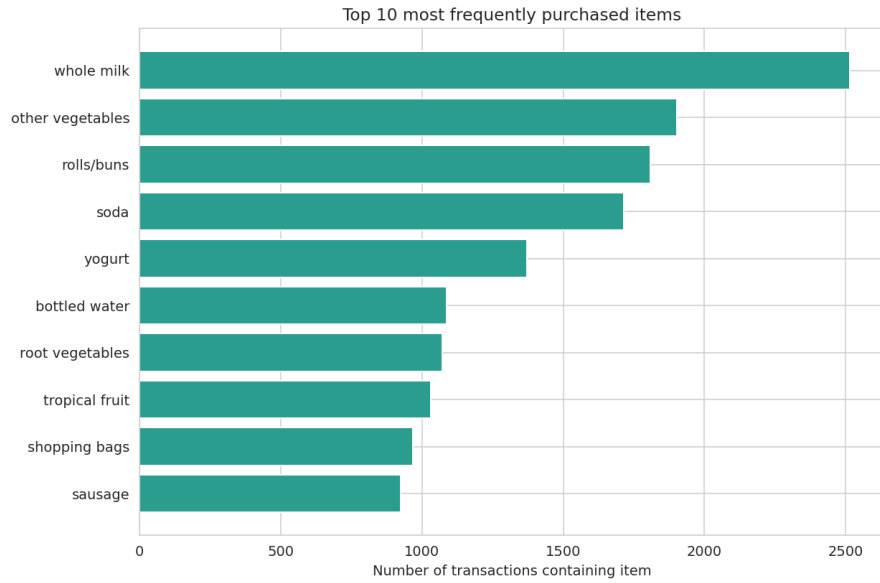


Figure 1: Top 10 most frequently purchased items.

## Question 2: Frequent Itemsets

Using the Apriori algorithm with a minimum support of 0.01 (1% of transactions), 245 frequent itemsets of 2 or more items were found. The top 15 (above the 10 required) are:

| Itemset                           | Support |
|-----------------------------------|---------|
| other vegetables, whole milk      | 0.0748  |
| rolls/buns, whole milk            | 0.0566  |
| whole milk, yogurt                | 0.0560  |
| root vegetables, whole milk       | 0.0489  |
| other vegetables, root vegetables | 0.0474  |
| other vegetables, yogurt          | 0.0434  |
| other vegetables, rolls/buns      | 0.0426  |
| tropical fruit, whole milk        | 0.0423  |
| soda, whole milk                  | 0.0401  |
| rolls/buns, soda                  | 0.0383  |
| other vegetables, tropical fruit  | 0.0359  |
| bottled water, whole milk         | 0.0344  |
| rolls/buns, yogurt                | 0.0344  |
| pastry, whole milk                | 0.0332  |
| other vegetables, soda            | 0.0327  |

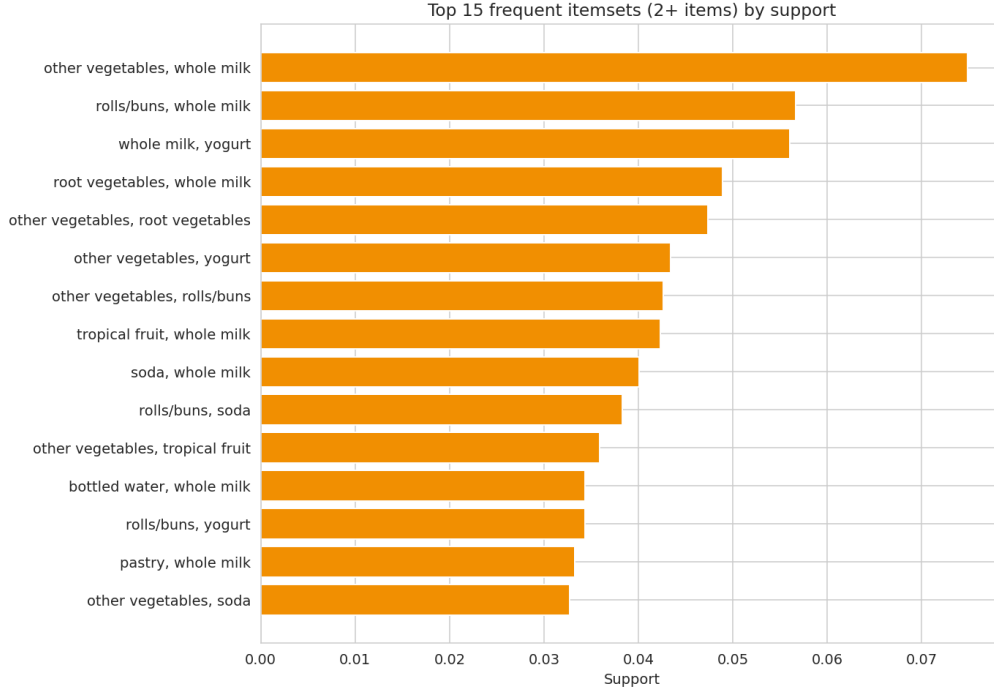


Figure 2: Top 15 frequent itemsets (2+ items) by support.

**c. Highest-support itemset:** {other vegetables, whole milk}, support = **0.0748** (appears in 7.48% of all transactions).

**d. What does support tell us?** Support measures how *common* a combination of items is across all shopping trips – it is simply the proportion of transactions containing that itemset. A high support value (like {other vegetables, whole milk} at 7.5%) tells us this is a routine, everyday combination for a large share of customers – consistent with both being staple grocery items bought on most shopping trips. Support alone says nothing about whether the items are *specifically* related to each other (that requires confidence and lift); it only reflects overall popularity.

### Question 3: Association Rules

Applying `mlxtend`'s association rule generator to the frequent itemsets (minimum confidence 0.2, keeping only rules with lift > 1) produced **231 rules**. The top 15 by confidence (above the 10 required):

| Antecedent                           | Consequent       | Support | Confidence | Lift  |
|--------------------------------------|------------------|---------|------------|-------|
| citrus fruit, root vegetables        | other vegetables | 0.0104  | 0.586      | 3.030 |
| root vegetables, tropical fruit      | other vegetables | 0.0123  | 0.585      | 3.021 |
| curd, yogurt                         | whole milk       | 0.0101  | 0.582      | 2.279 |
| butter, other vegetables             | whole milk       | 0.0115  | 0.574      | 2.245 |
| root vegetables, tropical fruit      | whole milk       | 0.0120  | 0.570      | 2.231 |
| root vegetables, yogurt              | whole milk       | 0.0145  | 0.563      | 2.203 |
| domestic eggs, other vegetables      | whole milk       | 0.0123  | 0.553      | 2.162 |
| whipped/sour cream, yogurt           | whole milk       | 0.0109  | 0.525      | 2.053 |
| rolls/buns, root vegetables          | whole milk       | 0.0127  | 0.523      | 2.047 |
| other vegetables, pip fruit          | whole milk       | 0.0135  | 0.518      | 2.025 |
| tropical fruit, yogurt               | whole milk       | 0.0151  | 0.517      | 2.025 |
| other vegetables, yogurt             | whole milk       | 0.0223  | 0.513      | 2.007 |
| other vegetables, whipped/sour cream | whole milk       | 0.0146  | 0.507      | 1.984 |
| rolls/buns, root vegetables          | other vegetables | 0.0122  | 0.502      | 2.595 |
| root vegetables, yogurt              | other vegetables | 0.0129  | 0.500      | 2.584 |

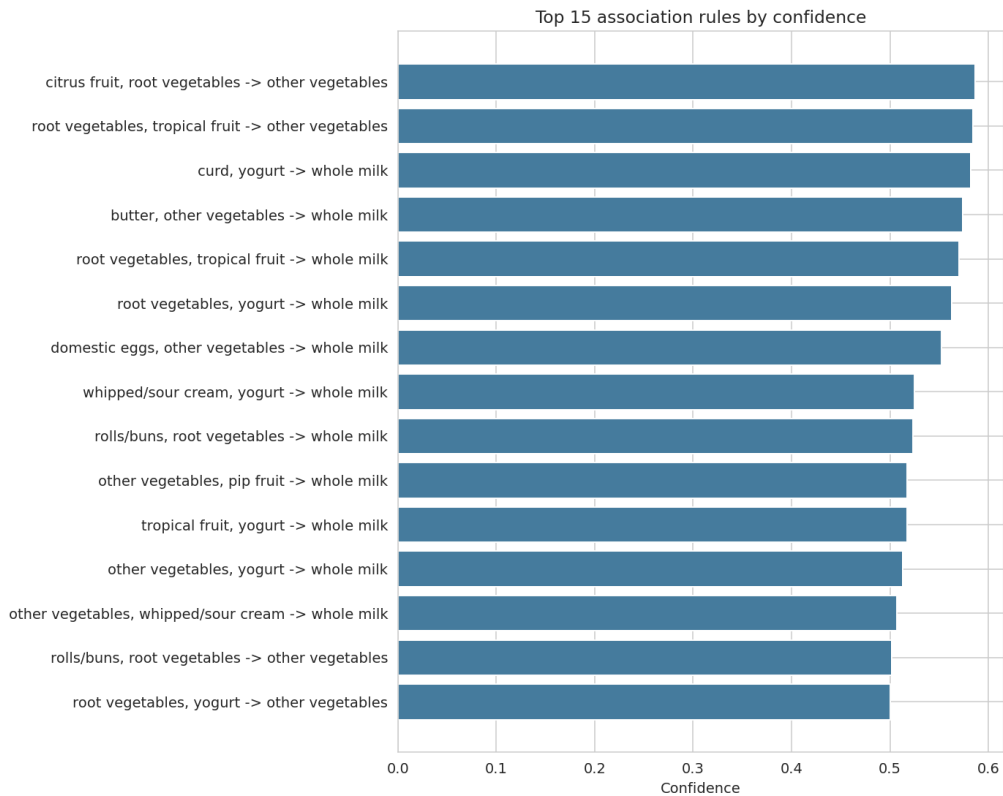


Figure 3: Top 15 association rules by confidence.



Figure 4: All association rules: support vs. confidence, colored by lift.

## Question 4: Understanding the Measures

Three example rules from the table above, explained:

1.  $\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$  (support 0.0104, confidence

0.586, lift 3.030)

- *Support:* This exact 3-item combination appears in about 1.0% of all 9,835 transactions.
- *Confidence:* Of all trips that included both citrus fruit and root vegetables, 58.6% also included other vegetables.
- *Lift:* Customers buying citrus fruit and root vegetables together are about 3 times more likely to also buy other vegetables than a random customer – a strong positive association.

**2. {root vegetables, tropical fruit} → {other vegetables}** (support 0.0123, confidence 0.585, lift 3.021)

- *Support:* This basket occurs in about 1.2% of all transactions.
- *Confidence:* 58.5% of trips with root vegetables and tropical fruit also included other vegetables.
- *Lift:* This combination makes buying other vegetables about 3 times more likely than chance – again a strong, non-coincidental relationship.

**3. {curd, yogurt} → {whole milk}** (support 0.0101, confidence 0.582, lift 2.279)

- *Support:* About 1.0% of all transactions contain curd, yogurt, and whole milk together.
- *Confidence:* 58.2% of trips with curd and yogurt also included whole milk.
- *Lift:* Buying curd and yogurt together makes buying whole milk about 2.3 times more likely than baseline – a moderately strong association, weaker than the produce-based rules above but still clearly positive.

**In general:** support = how common the whole rule is; confidence = how reliably the consequent follows the antecedent; lift = how much more likely the consequent becomes given the antecedent, versus its baseline popularity.

## Question 5: Find Interesting Rules

- **a. Highest confidence:** {citrus fruit, root vegetables} → {other vegetables} – confidence = 0.586
- **b. Highest lift:** {citrus fruit, other vegetables} → {root vegetables} – lift = 3.295
- **c. Highest support:** {other vegetables} → {whole milk} – support = 0.0748

**d. Are these the same rule?** No – all three are different. This is expected: the highest-*support* rule involves the two most popular individual items in the store (other vegetables, whole milk), which co-occur often simply because nearly everyone buys both regularly – yet the strength of their *specific* relationship is only moderate (confidence 0.39, lift 1.51). The highest-*confidence* and highest-*lift* rules, by contrast, involve more specific fresh-produce combinations that are comparatively rare in absolute terms (support  $\approx$  0.01) but represent a much stronger, more distinctive relationship once they occur. **Support rewards popularity; confidence and lift reward strength of association** – and the most popular items are not necessarily the ones most strongly linked to each other.

## Question 6: Interpret a Rule

**Selected rule:**  $\{\text{root vegetables, tropical fruit}\} \rightarrow \{\text{other vegetables}\}$  (support = 0.0123, confidence = 0.585, lift = 3.021)

**In plain language:** Customers who purchase root vegetables and tropical fruit in the same shopping trip also tend to purchase other vegetables. Out of every 100 customers who bought both root vegetables and tropical fruit, about 58 of them also bought other vegetables – roughly 3 times more often than would be expected if the items were purchased independently.

**Strong or weak association?** This is a **strong** association. The confidence (58.5%) is comfortably above 50%, so the consequent follows the antecedent more often than not, and the lift of 3.02 shows this isn't just because other vegetables happens to be popular on its own – the co-purchase pattern is genuinely stronger than chance. It is consistent with a recognizable “fresh produce / healthy shopping basket” behavior.

## Question 7: Business Application

**a. Two products to place close to each other:** *Root vegetables* and *other vegetables* – one of the strongest and most frequent 2-item pairings in the data (support  $\approx 0.047$ ), with a consistently high lift in both directions. Placing them in adjacent shelving would make it easier for shoppers already buying one to pick up the other.

**b. A possible product bundle:** A “**Fresh Produce Basket**” bundling *root vegetables* + *tropical fruit* + *other vegetables* – this is the exact combination behind the highest-confidence rule found (confidence 0.585, lift 3.02), so a bundle discount here targets a purchasing pattern that already occurs naturally and strongly among customers.

**c. Example recommendation use:** When a customer adds *root vegetables* and *tropical fruit* to their basket (in-store app or online cart), the system could automatically recommend *other vegetables* – e.g. “Customers who bought these items also bought other vegetables” – backed by the rule's 58.5% confidence and 3.02 lift, making it a data-justified suggestion rather than a generic upsell.

## Question 8: Understanding Association

**a. Does an association rule prove causation?** No. Association rules only show that items tend to be purchased together more often than chance would predict – a statement about *correlation*, not *causation*.  $\{\text{root vegetables, tropical fruit}\} \rightarrow \{\text{other vegetables}\}$  does not mean buying the first two *causes* the third purchase. A more likely explanation is a shared underlying cause: a shopper generally focused on fresh produce (e.g., for a healthy diet) tends to buy all three categories together. Association rule mining cannot rule out such confounding – results describe patterns, not causes.

**b. Why might a rule have high confidence but low lift?** This happens when the consequent is already very popular on its own, independent of the antecedent. If, for example, 80% of all transactions contain whole milk, almost any antecedent will show high confidence for  $\rightarrow \{\text{whole milk}\}$ , simply because milk is bought so often – but the lift, which compares that confidence to milk's overall frequency, may be close to 1 (or even below), revealing the antecedent isn't actually raising the odds beyond the baseline. High confidence with lift  $\approx 1$  is a rule that looks strong on the surface but mostly reflects the popularity of the consequent, not a real relationship.

**c. What does a lift value greater than 1 indicate?** A lift greater than 1 indicates a **positive association**: the antecedent increases the likelihood of the consequent beyond what independence would predict. The higher above 1, the stronger the relationship (lift = 3 means the consequent is 3 times more likely when the antecedent is present). Lift = 1 means no

association (statistical independence); lift  $< 1$  means a negative association – the items are bought together less often than chance, suggesting they may act as substitutes for one another.

## Summary

- 9,835 transactions, 169 unique items; purchases dominated by staple items (whole milk, vegetables, bread/rolls).
- 245 frequent 2+ item itemsets at 1% minimum support; the most popular pairing is {other vegetables, whole milk}.
- 231 association rules generated (support  $\geq 0.01$ , confidence  $\geq 0.2$ , lift  $> 1$ ); the strongest rules by confidence and lift both center on fresh-produce combinations (citrus fruit, root vegetables, tropical fruit, other vegetables).
- Support, confidence, and lift each answer a different question – popularity, reliability, and strength-beyond-chance, respectively – and which rule counts as “best” depends on which of these matters most for the business decision at hand.

*Full code and all generated tables/figures are provided in the accompanying Jupyter notebook (`association_rules_analysis.ipynb`).*